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Characterization and functional analysis of conserved non-coding sequences among poaceae: insights into gene regulation and phenotypic variation in maize

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Abstract

Background Conserved non-coding sequences (CNS) are islands of non-coding sequences conserved across species and play an important role in regulating the spatiotemporal expression of genes. Identification of CNS provides valuable information about potentially functional genomic elements, regulatory regions, and helps to gain insights into the genetic basis of crop agronomic traits.

Results Here, we comprehensively analyze CNS in maize, by comparing the genomes of maize inbred line B73 (*Zea mays* ssp. *mays*), its close wild relative *Zea mays* ssp. *mexicana*, and other grasses in Poaceae, including sorghum (*Sorghum bicolor*), foxtail millet (*Setaria italica*) and two adlay (*Coix lacryma*) cultivars. There were 289,931 CNS found in two syntenic gene pairs, while 51,701 CNS were conserved within at least three species. To explore the regulatory characteristics of the CNS identified, the flanking regions of CNS were compared with the peaks called using both transposase-accessible chromatin with high-throughput sequencing (ATAC-seq) and chromatin immunoprecipitation with high-throughput sequencing (ChIP-Seq) data of histone modifications. It was found that CNS in maize were enriched in open chromatin regions compared with randomly selected non-coding regions of similar length. A significant enrichment of transcription factor binding sites was found within CNS sequences, including different transcription factors involved in abiotic stress response, such as OBP (OBF-BINDING PROTEIN) family and Adof1 (Encodes dof zinc finger protein). To investigate the epigenetic modification patterns in CNS, ChIP-Seq data for histone modifications H3K9ac, H3K4me3, H3K36me3, H3K9me3, and H3K27ac were further analyzed to depict the changes along CNS. Our findings revealed significantly elevated levels of transcription-promoting histone modifications in the CNS regions compared to randomly selected non-coding sequences with an equal number and similar length. Notably, CNS were also identified on both *Vgt1* (Vegetative to generative transition 1) and *ZmCCT10*. In

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addition, CNS with potential functions were identified based on SNPs within CNS significantly associated with various agronomic traits in maize, which holds potential utility in molecular breeding for maize.

Conclusions In summary, we identified and characterized CNS in maize through genomic comparative analysis, which provides valuable insights into their potential regulatory effects on gene expression and phenotypic variation.

Keywords Conserved non-coding sequences, Maize, Regulatory Elements, Phenotypic variations

Introduction

Conserved non-coding sequences (CNS) are islands of non-coding DNA with over 70% sequence identity conserved across species for millions of years [1]. They typically range from 15 to 150 bp in length and are usually distributed in regions with low gene density [2], including introns, untranslated regions (UTRs), putative cis-regulatory elements, chromatin loop anchors, non-coding RNA genes, and certain transposon subfamilies [3]. Compared to protein-coding genes, our understanding of the non-coding sequences is limited.

With the advancements in high-throughput sequencing technologies and methodologies, more CNS have been identified and their functions in gene regulation, as well as the intricate regulatory networks involved, are being progressively unraveled [2–5]. CNS exhibits distinct characteristics including undergoing epigenetic modifications, being associated with open chromatin regions, and binding of transcription factors [1, 3, 6–8]. Histone modifications like H3K4me3, H3K9ac, and H3K4me2, which serve as markers of active enhancers [9–13], have been found enriched in CNS regions. In maize, a genome-wide approach called dCNS has been developed to find that CNS overlapped with H3K9ac modified regions [3]. In *Arabidopsis*, those histone modifications associated with transcriptional activation exhibit different signaling characteristics in CNS [12]. Additionally, a total of 855,366 CNS were discovered using rice as a reference to identify CNS across 50 well-annotated Gramineae genomes [5]. Interestingly, among these CNS, 37.04% were found to overlap with regions of open chromatin and transcription factor binding sites, indicating potential regulatory significance [5]. On the other hand, the histone modification H3K27me3 is retained after gene duplication and is correlated with conserved noncoding sequences in *Arabidopsis* [14].

Transcription factors regulate gene expression by binding cis-regulatory elements. CNS, often longer than typical transcription factor binding sites (6–12 bp) [15], frequently harbor overlapping binding sites for multiple transcription factors [1, 16]. For instance, CNS in the upstream region of the mouse Cluster of Differentiation 69 (*CD69*) gene contains binding sites for Octamer-binding transcription factor 1 (Oct1) and Runt-related transcription factor 1 (RUNX1). The expression of *CD69* is down-regulated when these transcription factors are

inhibited [17]. It is noteworthy that transcription factor binding sites are generally found in open chromatin regions, which frequently coincide with CNS locations [5]. In maize, the vegetative to generative transition 1 (Vgt1), located 70 Kb upstream of *ZmRAP2.7*, functions as a cis-regulatory element, repressing *ZmRAP2.7* expression and resulting in earlier flowering time [18, 19]. Several CNSs have been identified within the functional region of Vgt1, suggesting their potential involvement in the regulation of flowering in maize [3, 18]. Additionally, CNS in non-coding RNAs can also function as miRNA mimicry. For example, INDUCED BY PHOSPHATE STARVATION1 (*IPS1*) is a non-protein coding gene in *Arabidopsis thaliana*. It contains a conserved 23nt non-coding sequence, which is essential for miRNA-guided cleavage. The overexpression of *IPS1* leads to an increased accumulation of *PHO2* (encodes an E2 ubiquitin conjugase-related protein) mRNA, which is the target of miR-399, resulting in reduced phosphate content in plants [20, 21].

Despite occupying a larger genome portion than coding sequences, the identification and functional analysis of CNS among maize and its wild relatives remain underexplored. In this study, we aim to identify CNS in maize, by comparing the genomes of *Zea mays* B73, its close wild relative *Zea mays* spp. *mexicana* and other grasses in Poaceae, including sorghum, foxtail millet (*Setaria italica*) and two adlay (*Coix lacryma*) cultivars. Comprehensively analyses were performed to reveal the characteristics of CNS, including the overlaps of open chromatin, enrichment of transcription factors binding sites, and regions with epigenetic modifications (including histone modifications and DNA methylation). In addition, the potential functions of CNS were estimated by the number of variations in CNS that were significantly associated with maize root traits and different agronomic traits. These findings could highlight the potential role of CNS in regulating plant development and provide valuable insights for further applications in maize molecular breeding.

Results

Identification of CNS among maize and its relatives in Poaceae

A total of 143,326 collinear gene pairs between maize and the other five genomes were identified, involving

23,869 genes. Among the collinear gene pairs, we found that 123,167 gene pairs, accounting for 85.9% of the total, exhibited high collinearity between maize and all five other genomes, indicating a high ratio of conservation in the collinearity of these genes across different species. A total of 688,687 CNS were detected using STAG-CNS in all collinear gene pairs between maize and the other five genomes [22]. A total of 289,931 CNS were found in two syntenic gene pairs, while 51,701 CNS were conserved within at least three species. Notably, the number of CNS, amounting to 225,451, was present among all species detected (Fig. 1A). Further analysis of CNS length revealed that 95.4% of the CNS length was mainly concentrated within 50 bp, and mainly around 8–10 bp (Fig. 1B). We grouped the CNS based on the times at which combinations they were simultaneously detected, and we found that the shorter CNS length is more frequently detected, indicating stronger conservation across different genomes (Fig. 1C). To confirm the correlation between CNS length and its conservation, the length of CNS and the number of syntenic gene pairs was estimated and a negative correlation was observed by Pearson's correlation (correlation coefficient $R = -0.12$, $P < 2.2e-16$). It also suggests that CNS with stronger conservation tend to be shorter in length. For the distribution of CNS, we found that the majority of CNS (99.09%)

were located in intergenic and promoter regions, indicating their potential regulatory role outside of protein-coding regions (Fig. 1D). The distance of CNS to the nearest transcriptional start site (TSS) and transcription termination site (TTS) can provide insights into its regulatory potential and 391,150 CNS located more than 2 Kb away from the TSS and TTS (Fig. 1E).

CNS are enriched in open chromatin regions

To characterize open chromatin regions overlapping with the CNS, we generated 223,397,130 and 167,572,804 reads through ATAC-seq of maize inbred lines AC7643 roots under well-watered and water-stressed conditions, respectively. After reads qualification and mapping to the reference genome, 46,269 and 44,320 open chromatin regions were identified, respectively. The enhanced distribution of ATAC-seq reads in proximity to the genes serves as a strong indication of the successful establishment of the ATAC-seq library (Fig. 2A), which could contribute to effectively elucidating the open chromatin characteristics of the CNS.

Of the identified CNS sequences, 38,917 (5.65%) CNS were located in the open chromatin regions. To assess the significance of this finding, we selected 688,687 non-coding genomic fragments of similar length as the CNS, randomly sampled from the window in which the CNS

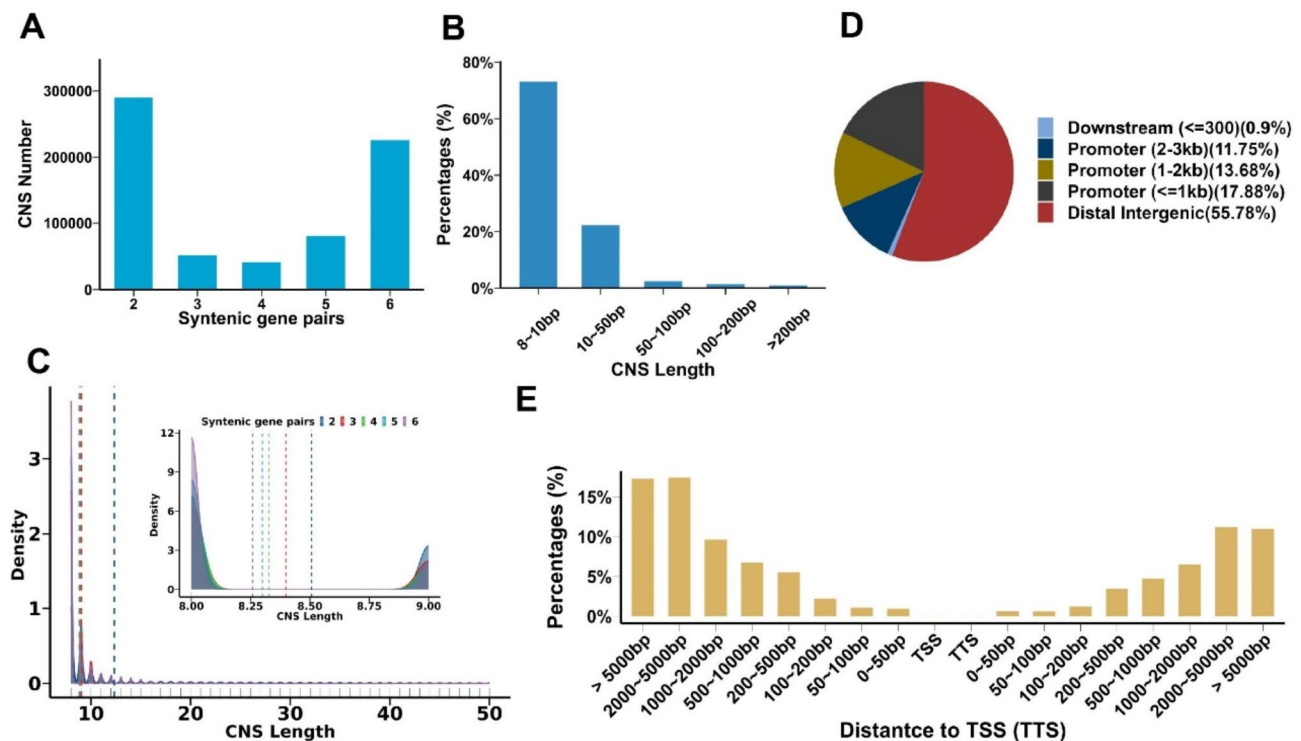


Fig. 1 Characteristics of CNS in Poaceae. **(A)** Frequency of CNS numbers derived from various collinear gene pairs among different genomes; **(B)** Percentages of the CNS with various CNS lengths; **(C)** Frequencies of CNS length derived from different syntenic genes (1–6 indicate Co-detection of CNS in 1–6 species); **(D)** Distribution of CNS in different gene structures; **(E)** Distribution of CNS in 5 Kb flanking sequences around gene transcription start site (TSS) and Transcription termination site (TTS)

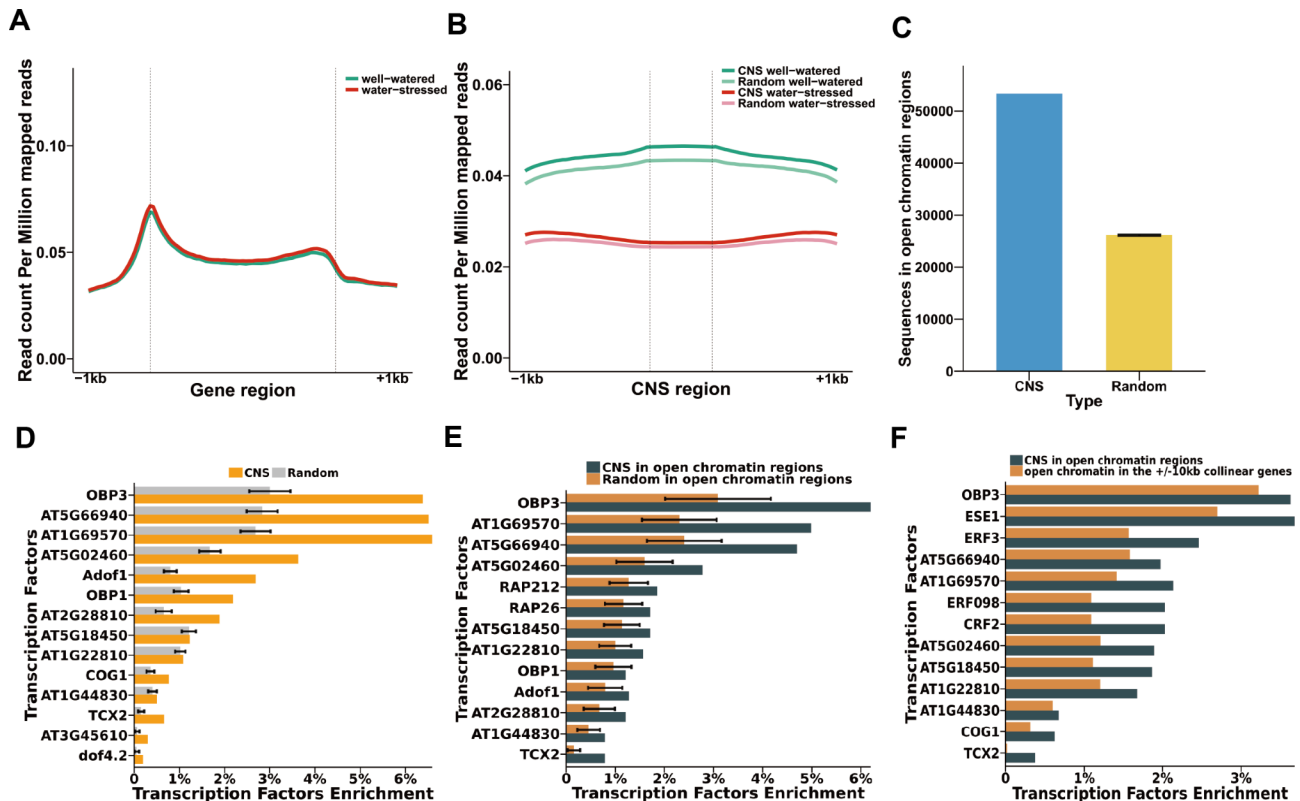


Fig. 2 CNS in maize are enriched in open chromatin regions and transcription factor binding sites. **(A)** The enrichment of ATAC-seq reads within the 1Kb region upstream/downstream of the gene. well-watered: ATAC-seq experiments under well watering conditions; water-stressed: ATAC-seq experiments under drought conditions. **(B)** Comparison of open chromatin levels between CNS sequences and randomly selected sequences. CNS: Conserved non-coding sequences; Random: non-coding genomic fragments of similar length as the CNS, randomly sampled from the window in which the CNS were identified. **(C)** Comparison of open chromatin levels between CNS sequences and randomly selected sequences in barplot. **(D)** The enrichment of transcription factor binding sites in CNS and randomly selected sequences, respectively. **(E)** The enrichment of transcription factor binding sites in CNS sequences that overlap with open chromatin regions and in randomly selected sequences (not overlapping with CNS) within open chromatin regions. **(F)** The enrichment of transcription factor binding sites in CNS sequences that overlap with open chromatin regions and in randomly selected sequences (not overlapping with CNS) in open chromatin regions within ± 10 kb of collinear genes

were identified, to serve as controls (1,000 bootstraps). The enrichment of open chromatin regions in CNS and their flanking sequences was estimated using the normalized ATAC-seq reads numbers. Compared with the randomly selected non-coding regions, CNS were shown significantly higher levels of open chromatin enrichment (Fig. 2B). The proportion of CNS and randomly selected control regions distributed in open chromatin was also examined and compared (Fig. 2C). It was observed that the distribution proportion of CNS in open chromatin regions was significantly larger than that of randomly selected sequences, confirming the enrichment of CNS in open chromatin regions (χ^2 test, $P < 2.2e-16$).

Enrichment of abiotic stress response-related transcription factor binding sites in CNS

To distinguish the special feature of CNS and its flanking sequences, DNA motifs recognized by transcription factors were identified using the FIMO tool from the MEME suite [23]. Significant enrichment of transcription

factor binding sites (TFBS) of the OBP (OBF Binding Protein) family and dof (DNA binding with one finger) family were found within CNS sequences (Fig. 2D). Also, we counted the number of CNS that contain TFBS and also overlapped with open chromatin regions. We compared the transcription factor enrichment in CNS and other random regions (not overlapping with CNS) in open chromatin regions in Fig. 2E. The regulatory motifs for transcription factors are enriched in CNS. We further compare the transcription factor enrichment in CNS and other random regions (not overlapping with CNS) in open chromatin regions within ± 10 kb of collinear genes, the regulatory motifs enrichment in CNS is also higher (Fig. 2F). Those TFBS enrichment in CNS overlapped with open chromatin regions was the highest among different categories (Fig. 2E and F), and the enriched transcription factor types found in our study show concordance with the findings in Arabidopsis [24], which suggests that CNS may play a crucial role in plant adaptation to environmental conditions.

DNA methylation and histone modification signatures around CNS

Modifications on DNA could impact gene expression by DNA methylation of neighboring TFBS or other regulatory elements. Thus, we plotted DNA methylation levels along CNS employing a sliding window approach (window size: 500 bp, step size: 100 bp) based on methylated DNA immunoprecipitation sequencing (MeDIP-Seq) data. The level of DNA methylation in CNS was depleted, compared to the flanking regions outside (Fig. 3A), which was in accordance with the findings before [25]. However, taking the randomly selected non-coding regions as a control, the DNA methylation level in CNS proximal (within 1 Kb) is significantly higher than control (Fig. 3A).

In addition, the functions of regulatory elements are determined not only by the DNA itself but also by epigenetic modifications presented on the encompassing histones. To investigate the epigenetic modification patterns around CNS, ChIP-Seq data for histone modifications H3K9ac, H3K4me3, H3K36me3, H3K9me3, and H3K27ac were further analyzed to depict the changes along CNS. As shown in Fig. 3B, CNS exhibited significantly higher levels of H3K9ac histone modifications compared to randomly selected sequences (randomly selected non-coding sequences with equal numbers and similar lengths as CNS, Fig. 3B). Although the CNS showed higher levels of H3K36me3 and H3K4me3 modifications compared to randomly selected sequences, the overall number of reads from immunoprecipitations was relatively low (Fig. 3C-D). Conversely, the level of H3K27ac modification was significantly lower in CNS compared to randomly selected sequences (Fig. 3E). The overall modification level of H3K9me3 along the CNS was relatively low (Fig. 3F). The levels of H3K27me3 modification along CNS sequences are similar with those of random sequences (Supplemental Fig. 1). In maize, the noncoding regulatory element *Vgt1*, known as a repressor of flowering, is situated 70 Kb upstream of *ZmRap2.7*. We found there were three CNSs enriched in *Vgt1*, overlapped with peaks of the enhancer marker H3K4me1 and transcriptional activation markers H3K9ac and H3K27ac (Fig. 3G). Notably, the absence of H3K27me3 peaks suggests a lack of repressive chromatin modifications. Additionally, we identified CNS located both upstream and downstream of *ZmCCT10* (Fig. 3H). The epigenetic modification patterns observed in these CNS are consistent with those of the *Vgt1* CNS (Fig. 3G). These results shed light on the epigenetic landscape of CNS and highlight their potential involvement in transcriptional regulation and gene expression modulation.

Functional potential of genes enriched for CNS

To explore the molecular functions of CNS-enriched genes and their evolutionary conservation, we categorize the CNSs involved genes into three groups: CNSs high-enriched ($n > 100$), CNSs low-enriched ($n = 1$) genes, and genes without CNSs. The majority of genes (76%) and their adjacent sequences (5 Kb) exhibited a count of fewer than 20 CNS, with an average of 15.68 CNS per gene (Fig. 4A). However, approximately 10% of the genes had more than 40 CNS in flanking regions (5 Kb), with 401 genes containing more than 100 CNS.

To clarify the differences in functions and evolutionary implications between genes possessing a substantial count of CNS and those exhibiting a lower count of CNS, we compared the maize-sorghum Ka/Ks ratio among the genes in the CNS-rich group, genes containing only one CNS (CNS-less group) and genes without CNS located in the 5 Kb flanking sequences. The comparison revealed that the Ka/Ks ratio of CNS-enriched genes was significantly lower than both that of genes containing only one CNS (Fig. 4B, Student's *t*-test, $P = 0.018$) and genes without CNS within 5 Kb proximity (Fig. 4B, Student's *t*-test, $P = 0.0056$). This indicates that CNS-enriched genes are more conserved.

To further investigate the potential functions of CNS-enriched genes, gene ontology (GO) enrichment analysis was performed to reveal that these genes are primarily involved in mRNA transcription (Biological Process) and oxidoreductase activity (Molecular Function), with incorporation or reduction of molecular oxygen and iron ion binding (Fig. 4C). The GO-enriched terms of genes with only one CNS or without CNS are similar, primarily enriched in Cellular Component, involving cellular components (Fig. 4C), which is different from the CNS-enriched genes. These findings provide valuable insights into the relationship between CNS abundance, gene conservation, and the functional roles of CNS-enriched genes.

To further analyze the functions of CNS, SNPs that overlapped with CNS were selected to estimate the correlations between CNS variations and agronomic traits, as well as variations and gene expression in maize [26, 27]. We identified a total of 6,709 SNPs that overlapped with CNS and were significantly associated with different phenotypical variations. In comparison, an average of 4,725 SNPs that overlapped with randomly selected fragments were found to be significantly associated with phenotypes. These random fragments were selected by matching the same number and similar length (excluding coding regions) from 10 kb upstream and downstream of the collinear genes with CNS identified (1,000 bootstraps). Four different groups are compared, including CNS in open chromatin regions, CNS not in open chromatin regions, randomly selected fragments in the +/-10 kb of

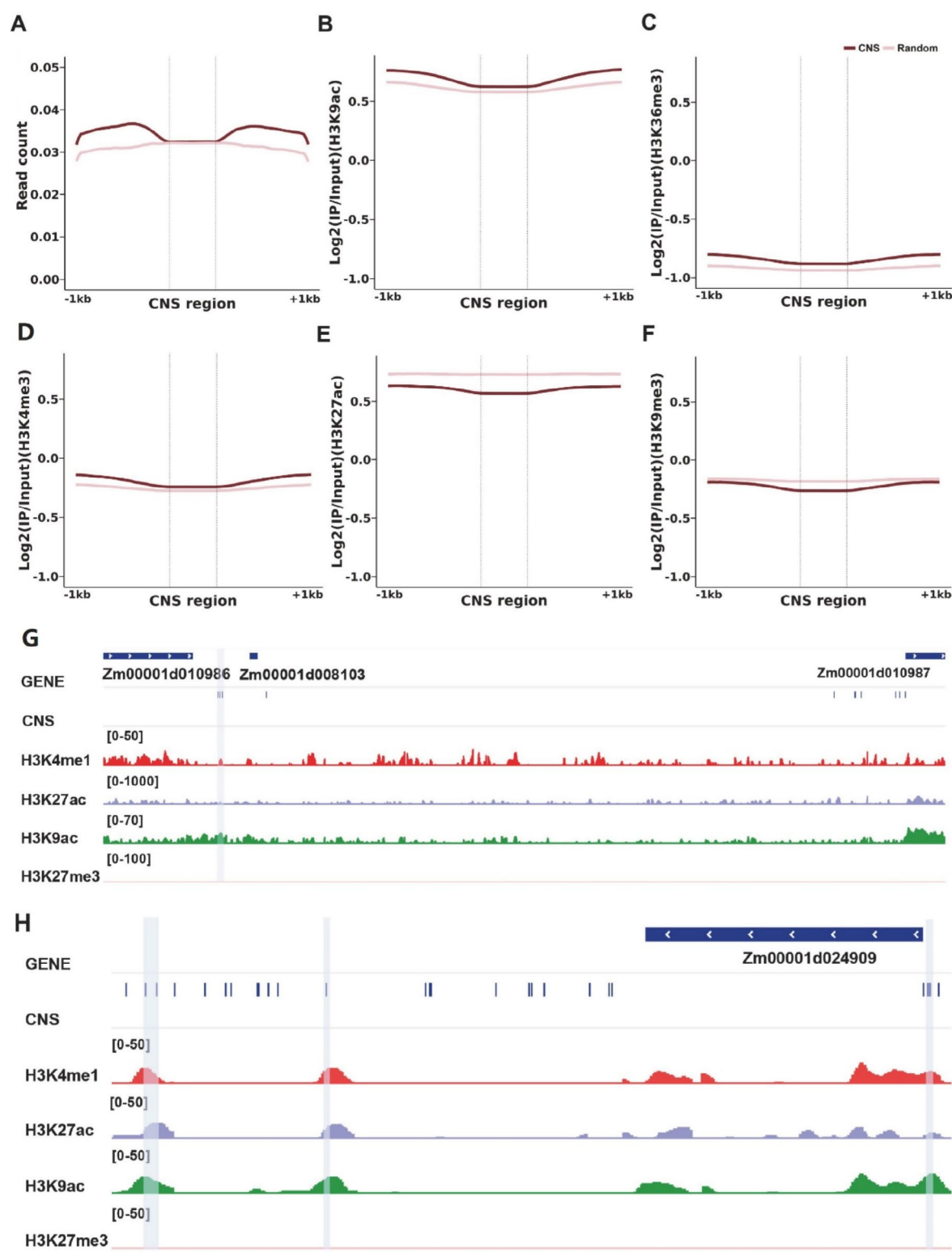


Fig. 3 DNA methylation and histone modification signatures around CNS. **(A)** Comparison of DNA methylation levels using MeDIP-Seq between CNS sequences and randomly selected sequences within the 1Kb regions upstream and downstream of the target sequences. Read count indicates the number of reads per million mapped reads. **(B)** Comparison of histone H3K9ac modification levels between CNS sequences and random sequences in the 1Kb region upstream/downstream of CNS sequences. **(C)** H3K36me3. **(D)**H3K4me3. **(E)** H3K27ac. **(F)** H3K9me3. **(G)** CNS within Vgt1 contains multiple histone modifications, including H3K9ac, H3K4me1, H3K27ac, and H3K27me3. **(H)** Multiple CNS surrounding Zm00001d024909 (*ZmCCT10*) exhibit a variety of histone modifications, including H3K9ac, H3K4me1, H3K27ac, and H3K27me3

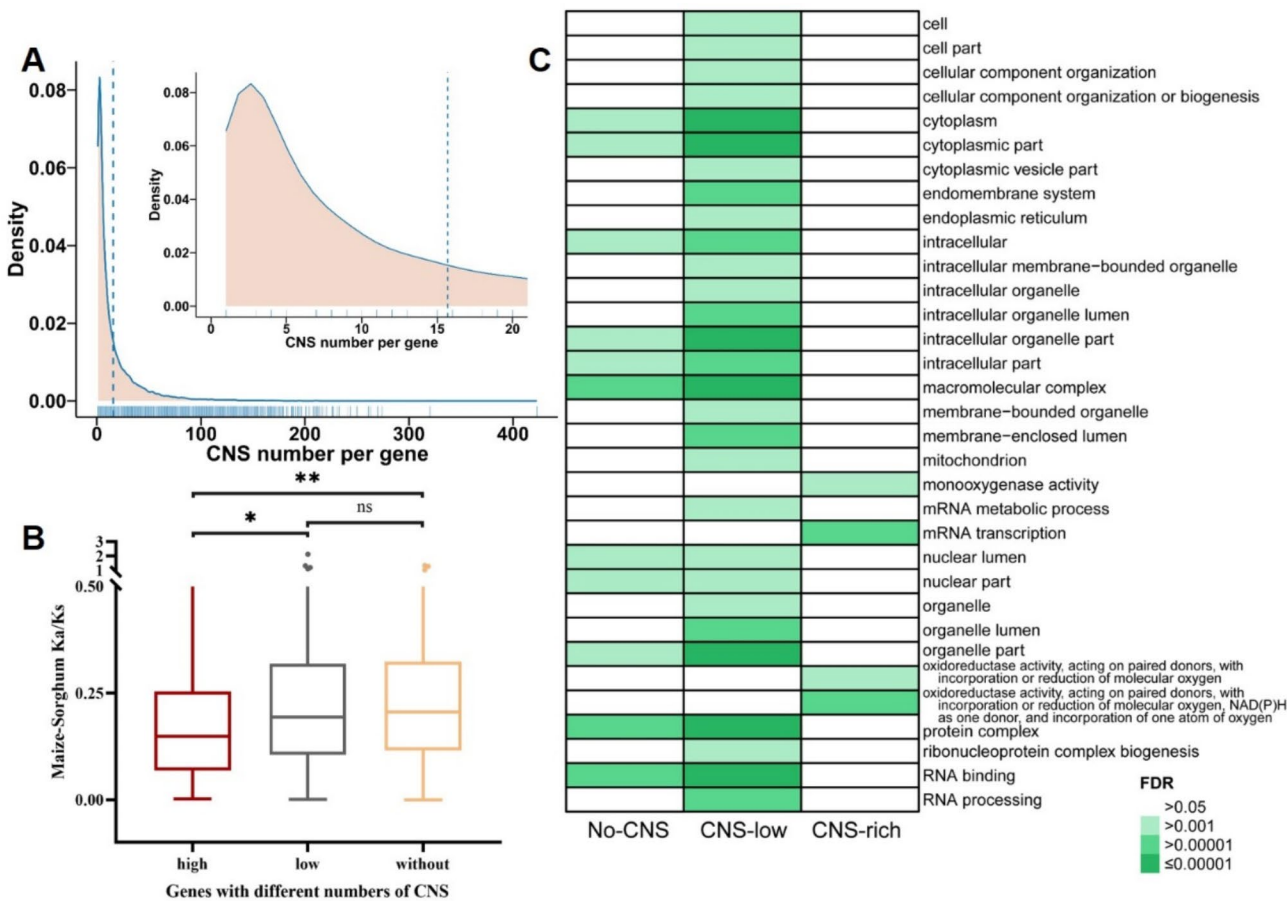


Fig. 4 Characteristics of genes enriched in CNS in maize. **(A)** Frequency of genes with different CNS numbers. **(B)** Comparison of maize-sorghum Ka/Ks ratios among genes with more than 100 CNS (high), genes with only one CNS (low), and genes without CNS (without) in the 5 Kb flanking sequences. Student's *t*-test, ns: $P > 0.05$; *: $P < 0.05$; **: $P < 0.01$. **(C)** GO enrichment analysis of genes with different numbers of CNS. No-CNS: indicates genes without CNS; CNS-low: genes with only one CNS; CNS-rich: genes with more than 100 CNS

collinear genes, and randomly selected fragments in open chromatin regions within the ± 10 kb of collinear genes. The number of SNPs in CNS that significantly associated with different agronomic traits was significantly larger than that in random fragments (Fig. 5A, Mixed linear model for GWAS, $P < 0.05$). We have integrated detailed information about the identified CNS, including their positions, neighboring genes, overlapping SNPs, and the traits with which SNPs were significantly associated, as well as open chromatin regions detected using ATAC-Seq, into Supplementary Table 1.

To explore whether variations within CNS are correlated with the expression of neighboring genes, we conducted an analysis of gene expressions across various tissues and developmental stages in maize, as well as gene expressions in the leaves at the seedling stage within a natural maize population under both well water and water-stressed conditions [28, 29]. A number of 4,588 SNPs located within CNS and significantly associated with any of the 17 agronomic traits (Fig. 5A) were

selected to investigate whether these CNS influence the expression of CNS-neighbouring genes. It was found that a total of 1,796 SNPs within CNS were correlated with the expression of CNS neighboring genes (1,167 genes). The GO enrichment analysis of these genes revealed that these genes play roles in several biological regulation processes (Fig. 5B), which provide insights into the molecular functions in which CNS might be involved.

In addition, we found 1,223 SNPs that not only affected the expression of CNS-neighbouring genes but were also significantly associated with agronomic traits (Student's *t*-test, $P < 0.01$). Among these loci, we focused on the gene encoding 40 S ribosomal protein (Zm00001d012886) and its flanking sequences (5 Kb), which contained 41 CNS. Notably, SNPs chr5.S_1568608 and chr5.S_1568631 were in CNS.391,063 and CNS.391,064, respectively. Analysis of gene expression of Zm00001d012886 in the natural population revealed significant differences among maize lines with different alleles (Fig. 5 CD, Student's *t*-test, $P < 0.01$), and the locus are also significant associations

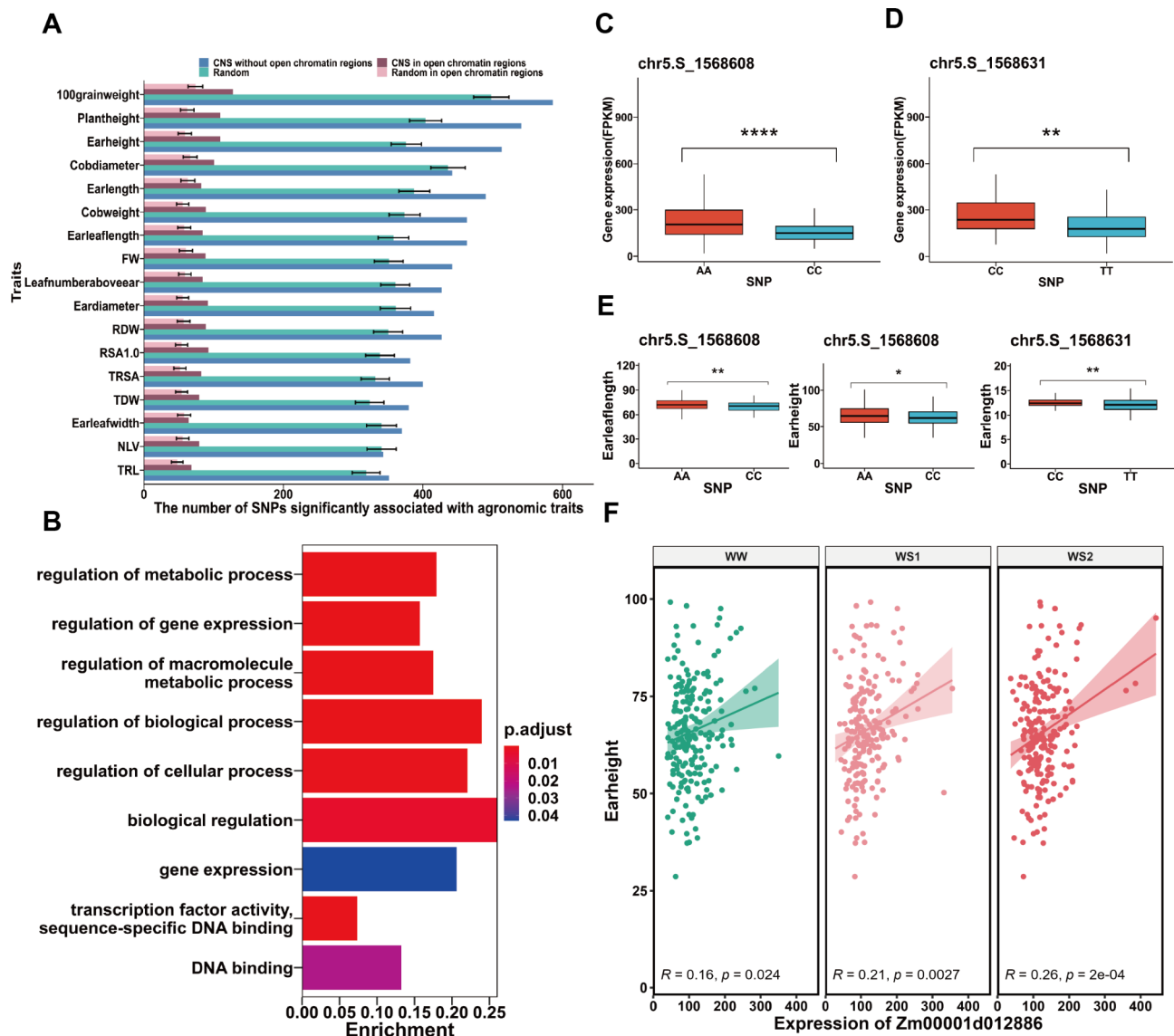


Fig. 5 The potential functions of CNS. **(A)** The number of SNPs significantly associated with different traits in CNS and randomly selected fragments. Four different groups are compared: CNS in open chromatin regions; CNS not in open chromatin regions; Random: randomly selected fragments in the ± 10 kb of collinear genes; Random in chromatin regions: randomly selected fragments in open chromatin regions within ± 10 kb of syntenic genes. FW: root fresh weight; RDW: root dry weight; TRSA: total root surface area; RSA1.0: root surface area at 1 mm root diameter; NLV: number of leaves; TRL: total root length; TDW: total dry weight. **(B)** GO enrichment analysis of CNS-neighbouring genes in different alleles. **(C)** SNP chr5.S_1608608 overlapping with CNS is correlated with the expression of Zm00001d012886 among the different allele types ($n = 360$). **(D)** SNP chr5.S_1608631 overlapping with CNS affects the expression of Zm00001d012886 among the different allele types ($n = 357$). **(E)** Significant differences in ear leaf length ($n = 480$), ear height ($n = 480$), and ear length ($n = 485$) between the different allele genotypes of the inbred lines. **(F)** Zm00001d012886 expression level and phenotype correlation analysis. WW: under normal water conditions ($n = 203$); WS1: mild drought ($n = 202$); WS2: severe drought ($n = 204$)

with ear leaf length and ear length (Fig. 5E, Student's *t*-test, $P < 0.05$).

Additionally, there was a significant correlation between the gene expression of Zm00001d012886 and ear height under normal water conditions (WW) (Pearson's correlation test, $P = 0.024$, $r = 0.16$). However, under moderate drought stress (WS1), a significant positive correlation was observed (Pearson's correlation test, $P = 0.0027$, $r = 0.21$). Under severe drought stress (WS2),

the positive correlation was even more significant (Pearson's correlation test, $P < 0.001$, $r = 0.26$) (Fig. 5F).

Discussion

CNS, which is not directly associated with protein-coding, often goes overlooked by researchers. However, compelling pieces of evidence support that CNS exhibit lower allele mutation frequencies due to purifying selection [1, 30–32]. The phenomenon was further proved by data

from a study of 1,000 human genomes [33]. In this study, we conducted a genome-wide identification of CNS and analyzed their potential biological functions. In maize, 99.09% of CNS were found in intergenic regions and promoter regions. CNS located closer to the TSS may have a more direct impact on gene expression, potentially acting as enhancers or promoters. Conversely, CNS located further away from the TSS may still have regulatory roles, such as long-range enhancers or insulators, influencing gene expression over larger genomic distances. This suggests that CNS acts in cis to coordinate target gene expression and ultimately affect the phenotype.

In eukaryotic cells, nuclear DNA is wrapped around histones in a structure known as chromatin, essential for DNA replication and gene transcription. For these processes to occur, chromatin must become accessible through histone modifications. When chromatin is opened, forming accessible regions, regulatory proteins such as transcription factors can bind to them, enabling the completion of replication or transcription. We found that the CNS overlapped well with the open chromatin regions in maize as revealed by ATAC-seq data (Fig. 2B), also supporting their roles in regulatory and in the form of other functional fragments (for example, enhancers, insulators, silencers). Additionally, the CNS identified is enriched with the histone modifications H3K9ac (Fig. 3), which promote transcript expression. It is worth noting that H3K9ac, H3K4me3, and H3K36me3 are known to be associated with transcriptional activation, while H3K9me3 is commonly found in heterochromatic regions and contributes to gene silencing [34, 35]. Conversely, the levels of repressor-associated modification H3K9me3 were significantly lower in CNS compared to randomly selected sequences. Chromatin undergoes dynamic changes, varying across different time points in diverse tissues and organs [36]. The CNS overlapping with H3K27ac signaling exhibited diverse trends of alterations across different organs in mice [37]. There were fewer H3K27ac signaling overlapping with CNS compared to H3K27ac signaling overlapping with randomly selected sequences in our findings, which may be due to tissue specificity. Both of the histone modifications H3K27ac and H3K4me1 serve as enhancer marks in animals. However, the histone modifications mark for enhancers are different between plants and animals. In rice, most enhancers do not colocalize with DNase I hypersensitive sites (DHSs) and lack the enhancer mark of histone modification H3K4me1 [38]. Additionally, enhancers are only part of CNS, for example, in vertebrate genomes, 75 out of 167 CNS, are enhancers [39].

Previous case studies in plants have demonstrated the functional significance of CNS in plant growth and development. Uchida et al. found two CNSs located in the proximal promoter of the Class 1 KNOTTED1-LIKE

HOMEBOX (*KNOX1*). One of these CNSs contains a known cis-acting binding motif called the K-box, while the other contains the RB-box. They elucidated the K-box, is critical regulatory element in the expression of SHOOT MERISTEMLESS (*STM*), *KNOX1* orthologous gene in Arabidopsis, which influences leaf development and potentially contributing to the diversity of leaf shapes in nature [40]. In previous maize GWAS results, the vast majority of molecular markers significantly associated with the phenotype were localized to the non-coding region, indicating the importance of those non-coding regions. Here, we provide evidence that CNS is significantly associated with gene expression regulation and phenotypic effects. Vgt1 is involved in the maize flowering regulatory process, and three CNS identified in our results are within this interval. The epigenetic modification component also predicts that these CNS may function as enhancers that increase gene transcription. Additionally, we found the presence of CNS enriched in both upstream and downstream of *ZmCCT10*, a gene known to play a significant role in maize flowering [41]. Notably, these CNS exhibited high levels of the H3K4me1, which is the histone modification marker of enhancers. The CNS located upstream is situated within the promoter of *ZmCCT10*. Furthermore, all of these CNS exhibit the transcription-promoting histone modifications H3K27ac and H3K9ac, suggesting their involvement in gene expression regulation.

CNS could be applied in crop breeding by regulating gene expression and ensuring the stability of genetic traits. These sequences, which do not code for proteins, include enhancers, promoters, and other regulatory elements that can influence the transcriptional activity of nearby genes. By understanding the function of CNS, researchers can identify key CNS linked regulatory networks that control desirable traits such as disease resistance, drought tolerance, and enhanced yield. The favorable CNS variations can serve as functional markers in crop molecular breeding. Furthermore, gene editing technologies, like CRISPR/Cas9, can be employed to modify specific CNS, which can help balance the trade-offs between various agronomic traits that often limit crop yield [42]. By targeting CNS, breeders can fine-tune the expression of key genes involved in traits such as grain size, drought tolerance, and disease resistance without the risks associated with altering protein-coding sequences. This precision enables the enhancement of multiple desirable traits simultaneously, addressing the challenges of achieving optimal crop performance. However, the limited number of natural functional variations in non-coding DNA poses a significant obstacle to crop breeding [43]. Therefore, the functional CNS identified in our study serve as valuable resources for molecular design breeding in maize, facilitating better adaptation

to changing environmental conditions and enhancing the ability to meet the growing demands for food security.

Methods

Identification of orthologous genes between species

In this study, we collected genome sequences and corresponding annotation files for NCBI (<https://www.ncbi.nlm.nih.gov/>) and Phytozome (<https://phytozome.jgi.doe.gov/pz/portal.html>), including the genomes of maize inbred line B73 (AGPv4), *Zea mays* spp. *mexicana* [44], sorghum (*Sorghum bicolor*) [45], foxtail millet (*Setaria italica*), two different *Coix lacryma* L. and perennial grass [46]. To identify collinear genes, MCscan in Tbtools [47] was applied to obtain collinear genome sequences and orthologous genes between maize and the other species.

Identification of CNS among species

The upstream 10 Kb and downstream 10 Kb sequences of each collinear gene pair were chosen for analysis. In cases where another collinear gene was located within this 10 Kb region, only the non-coding sequence between the target gene and the adjacent gene was extracted. Additionally, if a non-collinear gene was found within the 10 Kb region of a collinear gene, its non-coding sequence was categorized as a general non-coding sequence. STAG-CNS was used to identify the CNS based on the flanking sequences of collinear gene pairs [22]. R (4.3.0) was used for statistical analyses. The scripts for identifying CNS have been deposited on Figshare (https://figshare.com/articles/software/identifyPoaceaeCNS_zip/26499550?file=48190825).

ATAC-seq library preparation and data processing

Maize inbred line AC7643 was planted in a cultivation pool with a diameter of 40 cm and a height of 40 cm in Chengdu, China. The maize root cell nuclei at the 3-leaf stage of AC7643 were extracted and the ATAC-seq libraries were constructed according to the instructions of the TruePrep™ DNA Library Prep Kit V2 for Illumina (Vazyme, TD501). The quality of the libraries was assessed using the Agilent Bioanalyzer 2100 system. The qualified libraries were sequenced on the Illumina HiSeq-PE150 platform with a read length of 150 bp. FastQC [48] was used for reads quality assessment and clean reads were mapped to the maize reference genome (AGPv4) using Bowtie2 [49]. Only uniquely mapped reads were retained for peaks calling by MACS2 [50]. In addition, ATAC-seq sequencing data for maize inbred line B73 (SRR21230937 and SRR21230938) were downloaded and analyzed using the same pipeline. The peaks from different ATAC-seq data sets were merged using BEDTools [51]. The number of open chromatin regions overlapping with CNS was compared with that of the randomly sampled fragments (1,000 bootstraps). Initially,

we utilized the bedtools shuffle tool to randomly sample fragments from the genome. Following this, we excluded regions that overlapped with the coding regions of genes, while retaining non-coding regions that intersected with the flanking regions of CNS (upstream or downstream 10 kb). Subsequently, an equal number of fragments with similar lengths were randomly selected from previously chosen fragments to serve as the control group. Additionally, the BAM files were converted to BW files for visualization using deepTools [52].

We used the datasets of MeDIP-Seq and ChIP-Seq generated in our previous research, which can be accessed under accession number PRJNA294848 in the NCBI Sequence Read Archive and PRJCA018969 in the National Genomics Data Center [53]. Genomic regions with enriched AC7643 inbred lines root open chromatin were called using MACS2. All presentations of epigenetic features of CNS were computed using ngs.plot [54]. To facilitate the subsequent characterization of epigenetic features within CNS regions, the above sequencing data underwent conversion from the bam format to the bw format using deeptools [52].

Analysis of CNS-enriched transcription factors

The motifs of transcription factor binding sites were downloaded from the JASPAR motif database [55]. To identify specific DNA motifs around CNS that are recognized by transcription factors, the FIMO tool within the MEME suite [23] was used to analyze the CNS sequences and search for potential transcription factor binding sites. As controls, we employed a similar number of randomly selected sequences, each with the same length as the CNS sequences, extracted from 10 kb upstream and downstream of the CNS discovery genes to serve as controls (1,000 bootstraps). R was used to analyze statistical CNS transcription factor enrichment features.

Analysis of CNS characteristics

After CNS identification, the number and length of CNS identified in different species was calculated and compared using R-4.1.0. The distance between CNS to the nearest gene was calculated using R package ChIPseeker [56]. The ggplot2 package in R was used for data visualization [57]. The levels of various histone modifications (H3K9me3, H3K9ac, H3K27me3, H3K27ac, H3K4me1, H3K4me3, H3K36me3) and open chromatin level along CNS and randomly selected regions were calculated in ngs.plot [53, 54].

Identification of CNS significantly associated with different traits in maize

The results of association analysis with agronomic traits in maize are from our previous publications [26, 27]. To assess the potential functions of CNS, mixed linear

model in Tassel (version 3.0) was used to analyze the agronomic traits and root traits in a panel of 368 maize inbred lines as previously described [26, 27]. The positions of SNPs and CNS were compared using BEDTools (V2.26). The number of SNPs associated with different traits (Bonferroni-corrected P value < 0.01) was counted and compared among CNS and randomly selected sequences. AgriGO V2.0 [58] was used for GO analysis for CNS-neighbouring and CNS-enriched genes. Maize gene expression across various tissues and developmental stages, as well as gene expression in the leaves at the seedling stage within a natural maize population under both well-watered and water-stressed conditions, was obtained from previous research [28, 29]. All statistical procedures were calculated in the R-4.1.0.

Abbreviations

100grainweight	100-grain weight
Cobdiameter	Cob diameter
Cobweight	Cob weight
Pollenshed	Days to anthesis
Headingdate	Days to heading
Silkingtime	Days to silking
RDW	Dry Weight of Root
RSR	Dry Weight of Root / Dry Weight of Shoot
SDW	Dry Weight of Shoot
Eardiameter	Ear diameter
Earheight	Ear height
Earleaflength	Ear leaf length
Earleafwidth	Ear leaf width
Earelength	Ear length
Kernelnumberperrow	Kernel number per row
Kernelwidth	Kernel width
Leafnumberaboveear	Leaf number above ear
LRL	Longest Root Length
NLV	Number of Leaves in Visible
Plantheight	Plant height
RD	Root Diameter
RF	Root forks
FW	Root fresh weight
RL0.5	Root Length at 0.5–1 mm root diameter
RL0	Root Length at 0–0.5 mm root diameter
RL1.0	Root Length at 1–1.5 mm root diameter
RSA0.5	Root surface area at 0.5–1 mm root diameter
RSO	Root surface area at 0–0.5 mm root diameter
RS1.0	Root surface area at 1–1.5 mm root diameter
RT0.5	Root Tips at 0.5–1 mm root diameter
RT0	Root Tips at 0–0.5 mm root diameter
RT1.0	Root Tips at 1–1.5 mm root diameter
RV0.5	Root Volume at 0.5–1 mm root diameter
RV0	Root Volume at 0–0.5 mm root diameter
RV1.0	Root Volume at 1–1.5 mm root diameter
Tasselbranchnumber	Tassel branch number
Tasselmainaxislength	Tassel main axis length
TDW	Total Dry Weight
TRL	Total Root Length
TRSA	Total Root Surface Area
TRT	Total Root Tips
TRV	Total Root Volume

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12864-025-11221-9>.

Supplementary Material 2

Author contributions

JX designed the study. HZ, YL (Yi Luo), QW and YZ performed the data analysis. XZ, BY, TL and YM performed experiments for data collections. YW, YX and TL (Yaxi Liu) helped with libraries constructions. QT and YL (Yanli Lu) provided experimental materials. JX, YL (Yi Luo) and HZ wrote the manuscript. All authors contributed to the article and approved the submitted version.

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Data availability

The ATAC-Seq data is available at CNCB-NGDC under accession number CRA012202. The code for identifying conservative non-coding sequences is available on GitHub (<https://github.com/pccr-project/identifyPoaceaeCNS>).

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Conflict of interest

The authors have declared no Conflict of Interest.

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